

Start with a problem, not an idea

Dyson's Nathan Lawson McLean on making truly great products

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Few companies have managed to build a reputation in tech like Dyson. The brand is also synonymous with engineering-led innovation and thoughtful design. But what actually happens behind the scenes before a Dyson product reaches consumers? And where does the company see the future of home technology?

To discuss these points, I had a chance to sit down with Nathan Lawson McLean, Senior Design Engineer at Dyson, to understand how ideas evolve into products, the engineering challenges hidden inside seemingly simple machines, the role of AI and automation, and why Dyson believes there's still plenty of room for innovation in home cleaning.

Q Can you tell me how an idea is born at Dyson and what its product journey looks like?

A Nathan: Well, I'd say every product starts with a problem, not an idea first. We're always looking to solve real-world problems. One example is our dust illumination technology that you now see across most of our cordless vacuum range. It's all about revealing hidden dust around your home. That actually started inside our labs in the UK. Someone had brought in an enormous green laser after their child's party and noticed that particles in the air and smoke became illuminated when the laser was shone around.

We thought, 'What can we do with this?' We took a hot glue gun, attached the laser to an existing vacuum cleaner head, got down on the floor and started experimenting with different angles. When we found the right angle, it illuminated everything in front of it, including hidden dust.

From there, it goes through prototype after prototype before it leaves the lab. We show it to our chief engineers and founders, James and Jake Dyson, and then it gathers momentum. We create formal prototypes and take them through testing.

Q A Dyson vacuum or purifier looks simple, what's one piece of engineering inside that's complex?

A Nathan: There are many components inside a Dyson purifier. You've got the Air Multiplier technology, the impeller and the filter. The impeller is important because it's attached to the motor and is responsible for drawing air in and pushing it through the filtration system and Air Multiplier.

Noise is becoming more important than ever in people's homes, alongside filtration performance. It's a balancing act. You want strong performance without drawing too much power or generating excessive noise.

The motors inside our fans, and particularly the impellers, are incred-

ibly carefully engineered. We use computational fluid dynamics across our global development centres to constantly refine them.

Today, AI is helping us run thousands of simulations, allowing us to maximise performance, minimise acoustics and ensure products are suitable for home environments.

Q How do you see future homes evolving, and how do you make sure these technologies remain useful and not complicated?

A Nathan: Exactly. It's about solving real problems and being useful, not simply adding complexity. At Dyson, we've been devel-

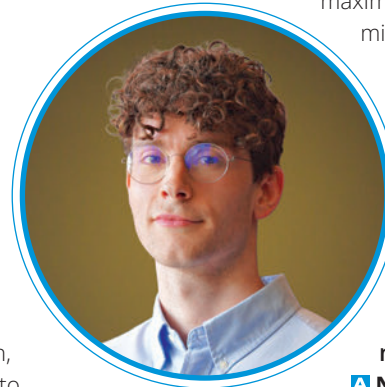
oping perception systems for around 30 years, going back to our earliest robotic vacuum concepts.

I like the word 'perception' because it's about understanding the home and user intent. What do users want to achieve? What's an intuitive way to interact with products? How products react and predict what users need?

You'll continue to see more of that. Previously, we used sensors primarily on robots such as the Dyson 360 Eye. Then we introduced Time-of-Flight sensors on products like the Super-sonic Nural to measure proximity and protect scalp health.

More recently, our latest purifiers include camera systems that can intelligently track users around the home.

We're excited about what perception, automation and predictive analysis can do, both in existing categories and new categories yet to come. **d**



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